

Ref No.: BD-CW-SAMCO-ACE-LET-STR-034

17-December-2025

To: ACE Project Management | Moharram Bakhoun

Attention: Eng. Abdulaziz Mohamed (Project Manager)

Project Name: The Big Business District.

Contract/Package: Construction Works.

Subject: Re: design review

Dear Sir,

With reference to the design review comments received from El Nile Engineering Consulting Bureau, we would like to clarify the following:

According to the loads, assumptions, and design criteria adopted by the Owner's Designer (Prime Consulting Office), and as coordinated with the General Consultant and Aspire Office as the Specialized Structural Consultant, there is no objection to commencing the execution works based on the issued design.

However, it is noted that El Nile Engineering Consulting Bureau has conducted its design review based on revised and more conservative design criteria, including, for example and not limited to, the following:

Adopting an increased seismic importance factor ($\gamma = 1.20$), Increasing certain dead loads, such as wall loads from 17.0 kN/m to 17.9 kN/m, and internal area finishes from 2.0 kN/m² to 2.5 kN/m², Increasing parking slab loads from 2.0 kN/m² to 2.5 kN/m², Applying additional plaster loads to basement slabs of approximately 4.2 kN/m²,

Revising the equivalent live load participation in seismic analysis from 25% to 50%, and modifying the associated load combinations accordingly.

Based on the results of the revised analysis under the above-mentioned criteria, El Nile Office concluded that the current structural design does not satisfy such revised criteria and therefore requires redesign.

Accordingly, and in order to avoid any misunderstanding, we confirm that:

Execution can proceed if the project adopts the design criteria and load assumptions of Prime Consulting Office, and

In the event that the project elects to adopt the criteria and assessment approach of El Nile Engineering Consulting Bureau, a full structural redesign will be required prior to execution.

Accordingly, the matter is respectfully left to your discretion to take whatever action you deem appropriate.

Best regards,

Eng./ Mohammed Sami Mohammed
Project Director
SAMCO National Construction



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Project Director
SAMCO National Construction



PROJECT: THE BIG BUSINESS DISTRICT New Cairo, Cairo

OWNER: ROYA DEVELOPMENTS



CONSULTANT: PRIME CONSULTING OFFICE



CONTRACTOR : SAMCO



SAMCO National Construction Co.
سامكو الوطنية للتشييد

CONTRACTOR CONSULTANT : NECB



BUILDING (B1 & B2) Structural design check report

Rev : 00

Date : 17/12/2025





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1. Introduction

This report presents the structural design check for **Buildings B1 and B2** within The Big Business District project.

2. Building Description

- **The part consists of** two reinforced concrete buildings (**B1 and B2**) that are connected at the foundation, Basement 01, and Ground Floor levels.
- **The** two buildings differ in the number of stories **above ground**.
- **The** structural system **comprises** a flat slab with drop panels, **supported by** reinforced concrete columns.
- **The** foundation system **consists of** a raft foundation, **which is** shared with the adjacent Buildings B3 and B4.
- **The building (B4) include** underground water tanks, **which also** support the vertical loads of the superstructure columns.

3. Objectives of the Design Check

The purpose of this design check is to ensure that the structural design of Buildings B1 and B2 satisfies the following criteria:

- **Documentation and Approval:** Proper documentation, traceability, and adherence to the established approval process.
- **Compliance:** Conformance with all **applicable codes, standards, and project specifications**.
- **Performance:** Assurance of **safety, serviceability, and constructability** under all design load conditions.





4. Available Documents

- Architectural drawings (ISSUED#27-10-2025)
- Structural drawings (ISSUED#27-10-2025_Not updated)
- Structural models (ISSUED#7-12-2025)
- Structural calculation report (ISSUED#27-10-2025_Not updated)

5. Reference Standards

- ISO 9001:2015 – Quality Management Systems
- ECOP (201-2012) EGYPTIAN CODE FOR CALCULATION OF LOADS AND FORCES ON BUILDINGS.
- ECOP (203-2020) EGYPTIAN CODE FOR DESIGN AND CONSTRUCTION OF CONCRETE STRUCTURE.
- ECOP (202-2005) EGYPTIAN CODE FOR SOIL MECHANICS, DESIGN AND CONSTRUCTION OF FOUNDATIONS.

6. Design criteria

- After reviewing the general note drawings and the calculation report on page 12, some comments were found:

A- Dead loads

1- Block work

- The block work consists of hollow cement blocks for the main wall construction.
- Solid cement blocks are used for the first 1.00 meter above the floor level, around wall openings, and at every 3.00 meters of wall height to enhance wall strength, stability, and impact resistance.
- Plaster Load:
The plaster is considered on both sides of the wall, with an estimated unit weight of: $0.05 \text{ m (thickness)} \times 2100 \text{ kg/m}^3 = 105 \text{ kg/m}^2$
- Wall Load adjustment:

The total wall load used in the structural model increased to be (17.90 kn/m) instead of (17 kn/m) and for internal open spaces increased to be (2.50 kn/m²) instead of (2.00 kn/m²) as necessary to match the actual wall and finish loads, ensuring realistic representation of self-weight and imposed loads in the design.





2- Parking pavement

- As per the coordination meeting ,the pavement material is foam concrete slab with thickness 15 cm with an uniform unit weight = 2.50 kn/m^2
- The pavement loads adjustment:

The pavement loads used in the structural model increased to be (2.50 kn/m^2) instead of (2.00 kn/m^2) as necessary to match the actual loads.

3- Plaster loads

- According to the architectural drawings, Basement slabs include plaster on the underside. The plaster is considered 2.00 cm estimated uniform load = $0.02 \times 2100 \text{ kg/m}^3 = 42 \text{ kg/m}^2$
- The plaster loads adjustment:
- The plaster loads used in the structural model added to be (4.20 kn/m^2) as necessary to match the actual loads.**

B- Live loads

1- Parking access height more than 2.40m

- As per ECP 201 – table 4-1 = 400 kg/m^2
- The parking loads adjustment:

The parking loads used in the structural model increased as mentioned to match the actual loads.

C- Seismic Loads

1- As per ECP 201 – page 8-9 the importance Factor $V = 1.20$ – **(Modified)**

2- As per ECP 201 – page 8-7 the equivalent Live load percentage = 50% **(Modified)**

D- Temperature Loads

3- As per ECP 201 – 20 degree of temperature change should be applied for the floors more than 45.00 m length – **(Modified)**





7. Documents comparing

- After reviewing the Structural drawings and the architectural drawings ,some comments were found: (see attached drawings)

8. Structural models input review

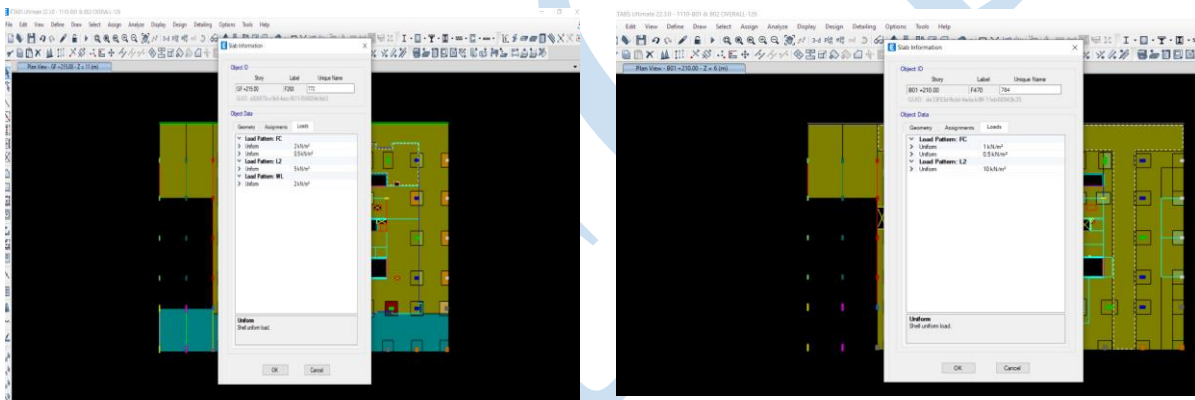
- After reviewing the Structural drawings and the models ,some comments were found: (see attached drawings)

A- Geometry

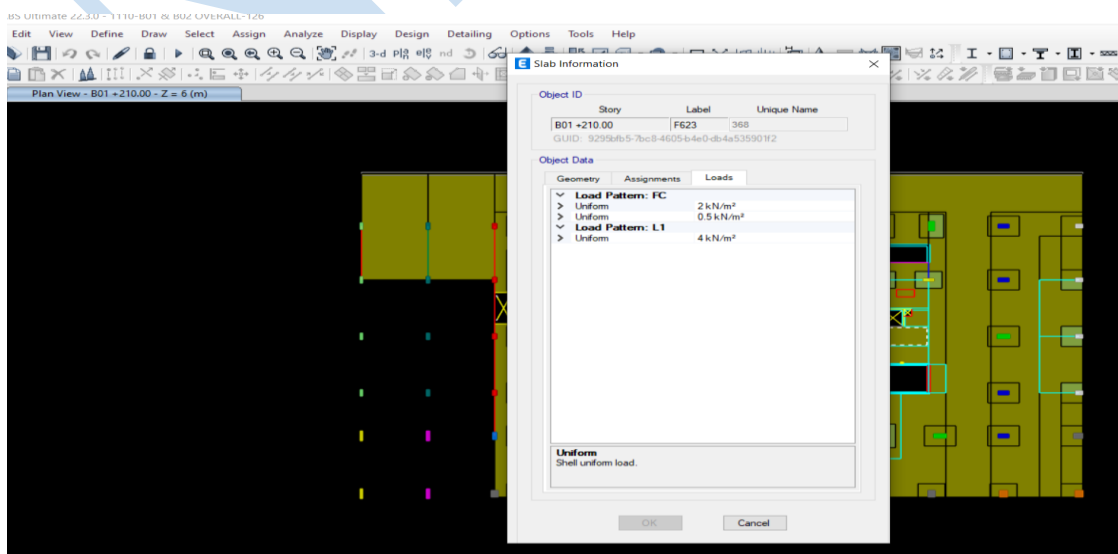
- The columns sections and reinforcement in the model bigger than the drawings for 19 columns.
- The total length of B1 in the model is 0.40 m shorter than shown in the drawings from level 237.90 to 248.80.

B- Etabs Model Input

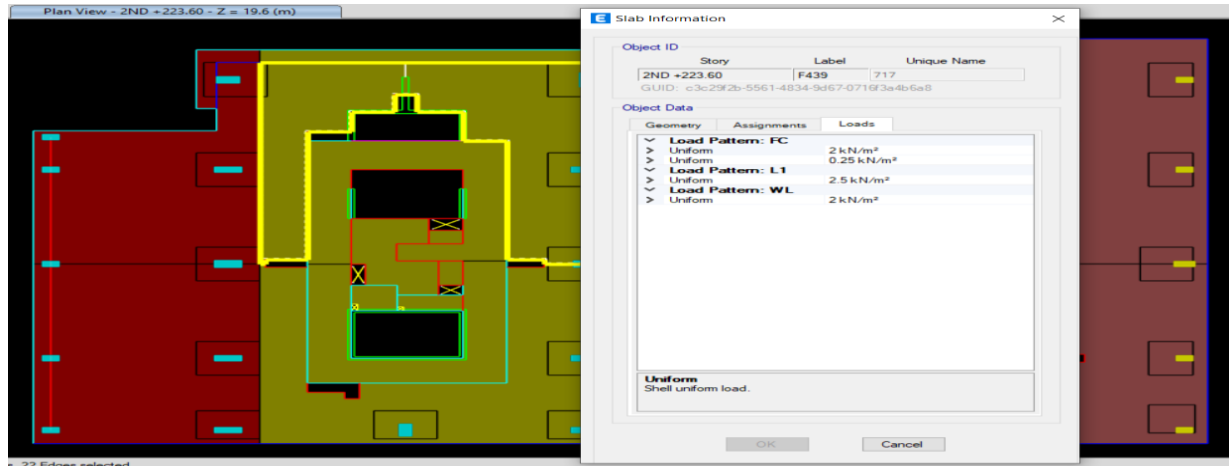
- The plaster loads is missing for Basement slabs.



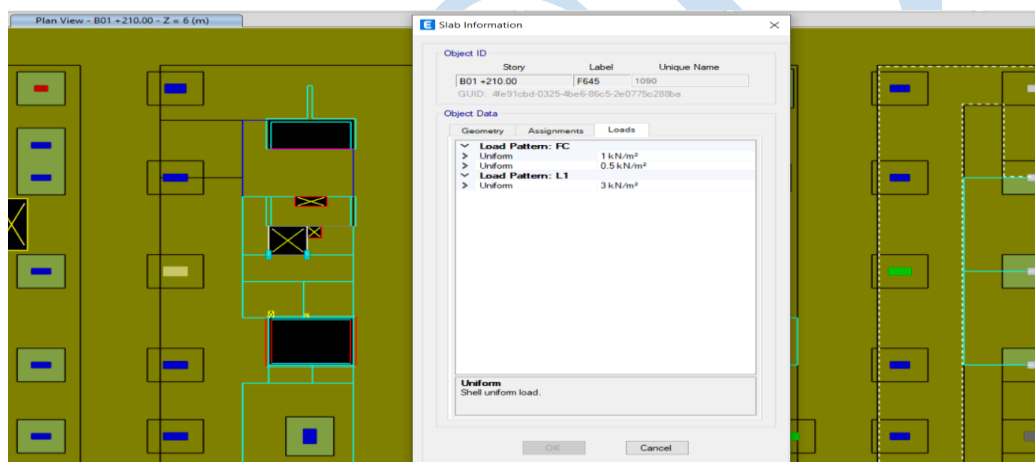
- The pavement loads need to increase.



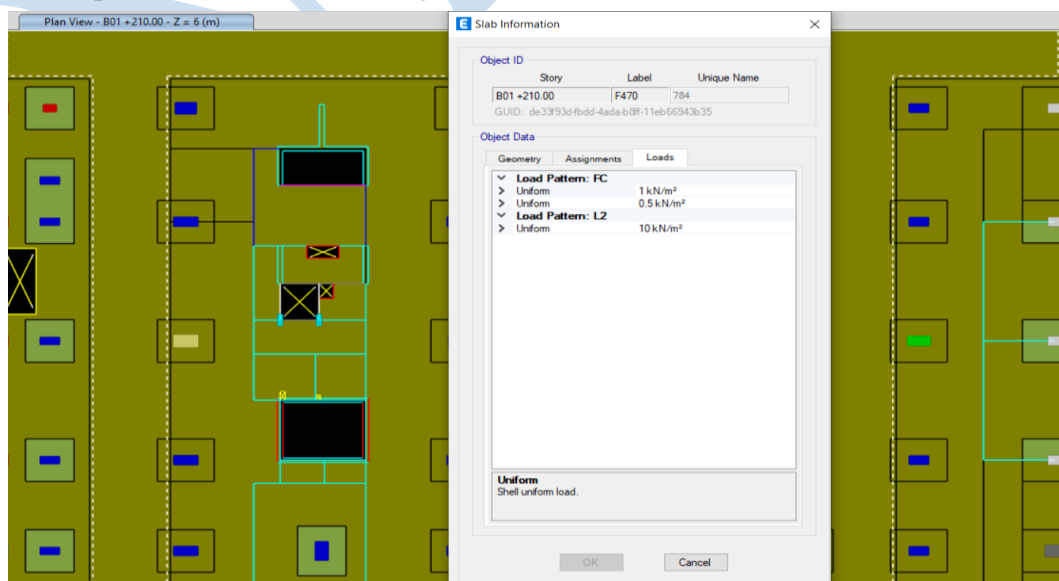
3- The block wall loads need to increase for all floors.



4- The parking area live loads need to increase.



5- The temperture load is missing.





6- As per ECOP-201, Clause 2-1-2-1, the live load should be included only once in the load combination (LL).

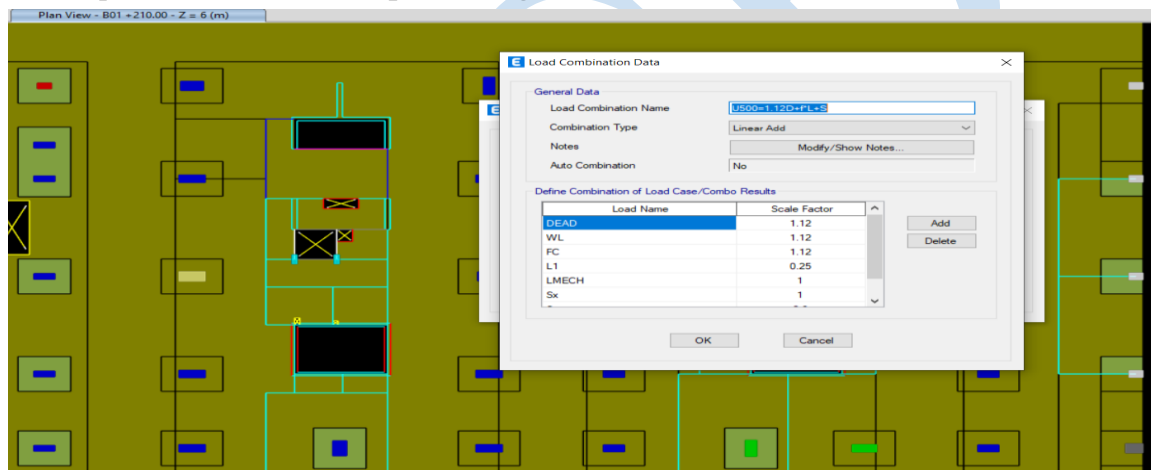
٢-١-٢ الأحمال والقوى الحية

١-٢-١-٢ الأحمال الحية

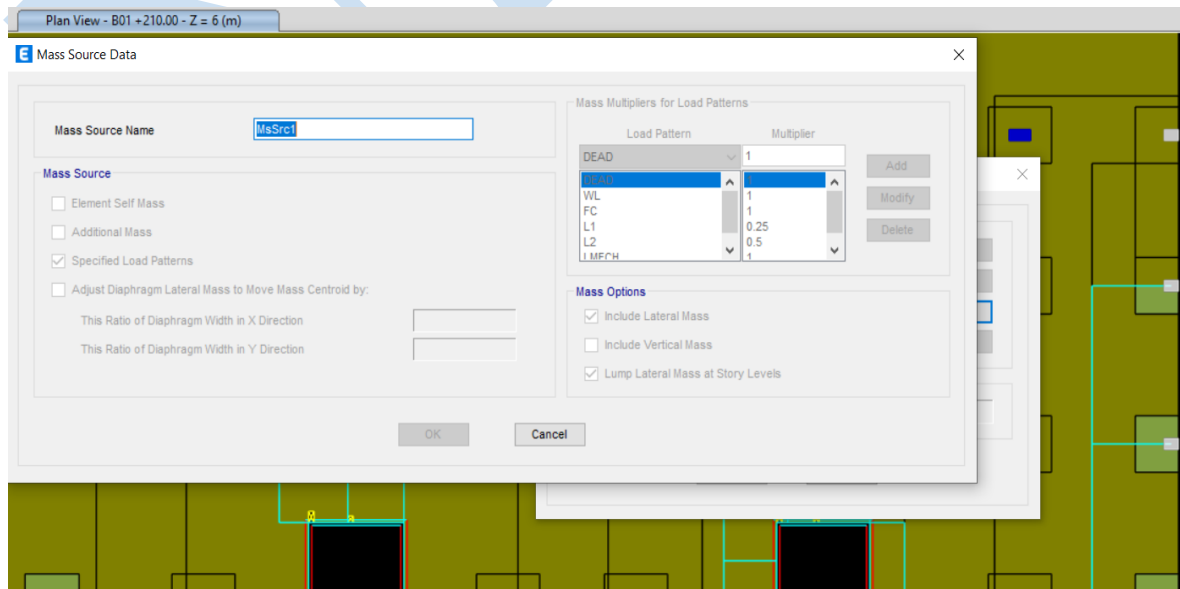
وهي الأحمال المتغيرة والمتحركة التي يتعرض لها أى جزء من المنشأ بما فى ذلك الأحمال الموزعة والمركزة وأحمال الصدم والاهتزازات والقصور الذاتي وهي تشمل:

- أ - أوزان الأشخاص مستعملى المنشأ.
- ب - أحمال الماكينات وإهتزازاتها.
- ج - أحمال الأثاث والأجهزة والآلات غير المثبتة ومواد التخزين.

7- The equivalent Live load percentage = 50% instead of 25 %



8- The live load factor at mass source = 0.50 instead of 0.25

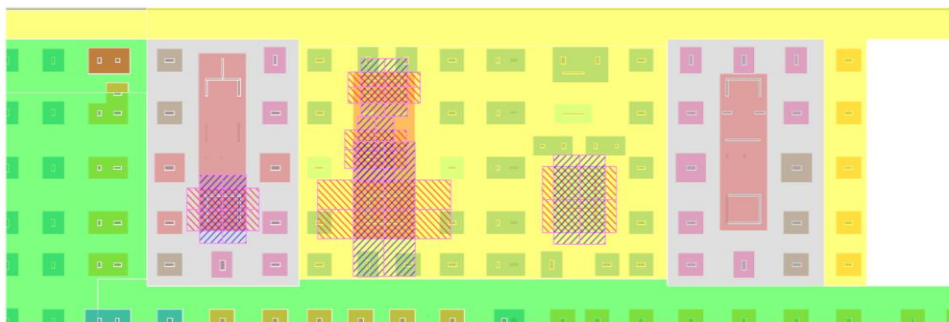




9- As per ECP 201 – page 8-9 the importance Factor ($\gamma = 1.20$).

C- Safe Model Input

1- The live loads is missing.



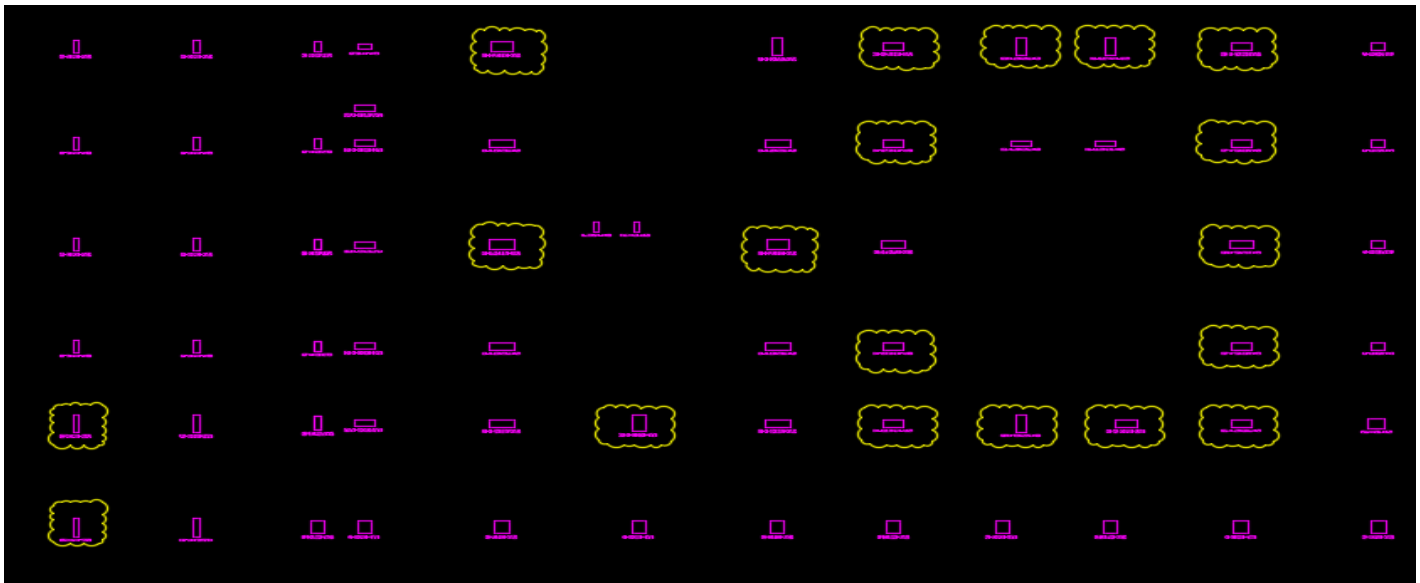
2- The foundation should be designed considering the seismic loads from the ETABS model, after modifying the response reduction factor (R) to 3, in accordance with ECOP-201, Clause 8-8-2-6



9. Structural models output review

- After made all above modification and run the Structural 3d model ,the vertical elemnts results as followed:

A- Columns



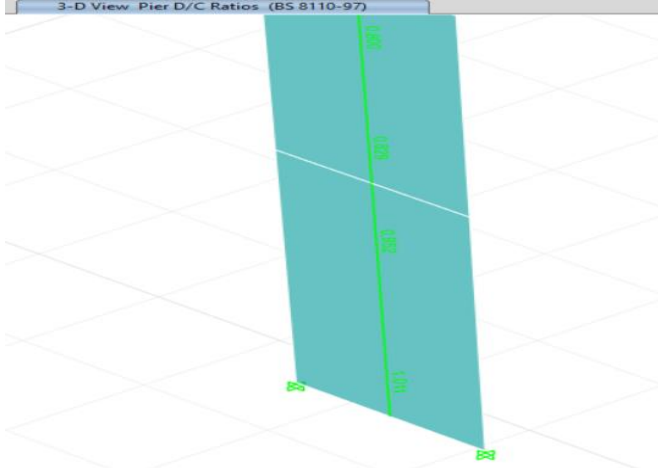
The columns sections and reinforcement in the model bigger than the drawings for the clouded 19 columns .

Plan View - B01 +210.00 - Z = 6 (m) Column P-M-M Interaction Ratios (BS 8110-97)

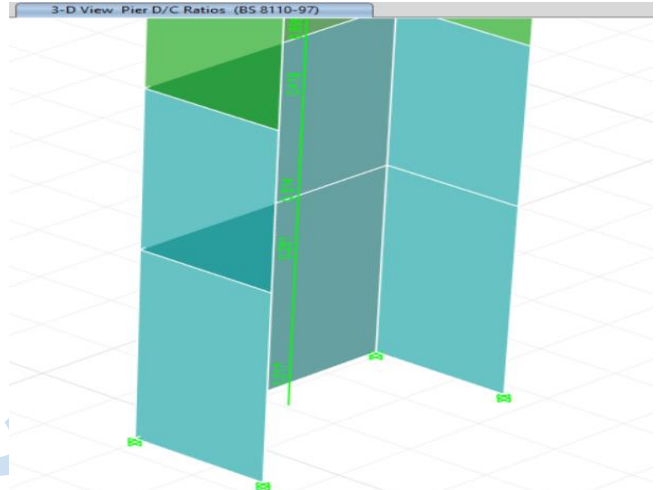


The mentioned columns columns are consider unsafe + the above columns.

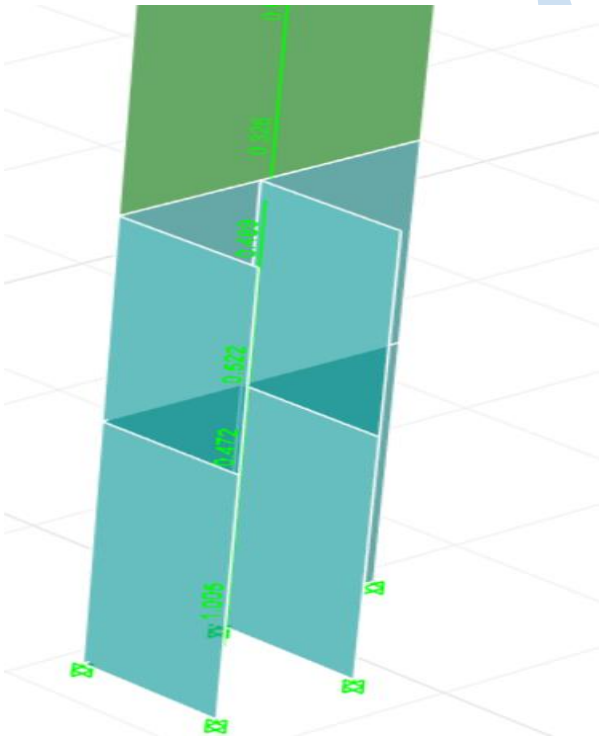
B- Shear walls



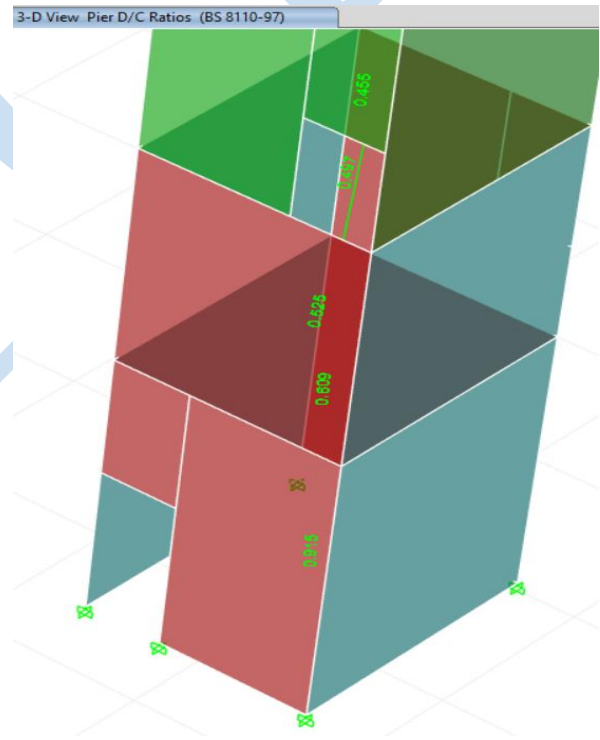
Wall – B01-W2



Wall – B02-W1



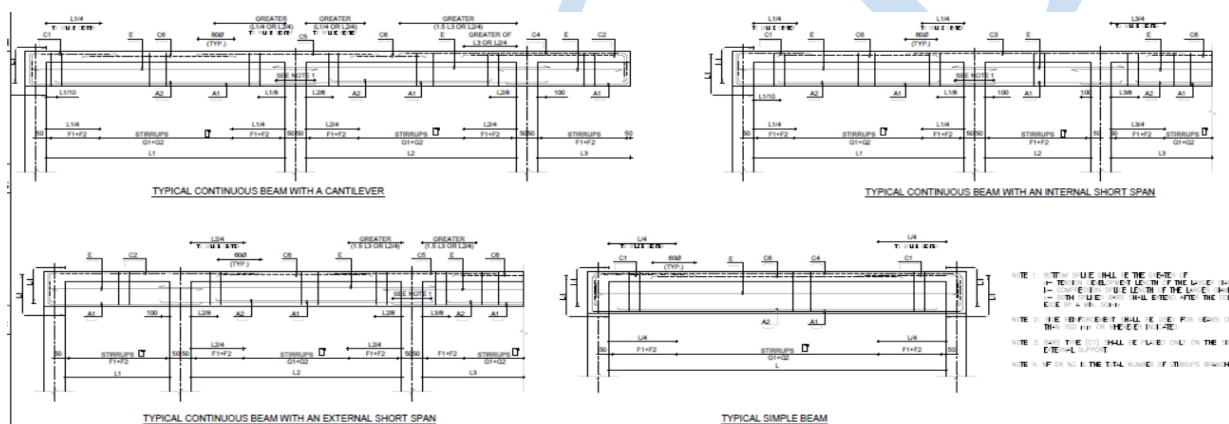
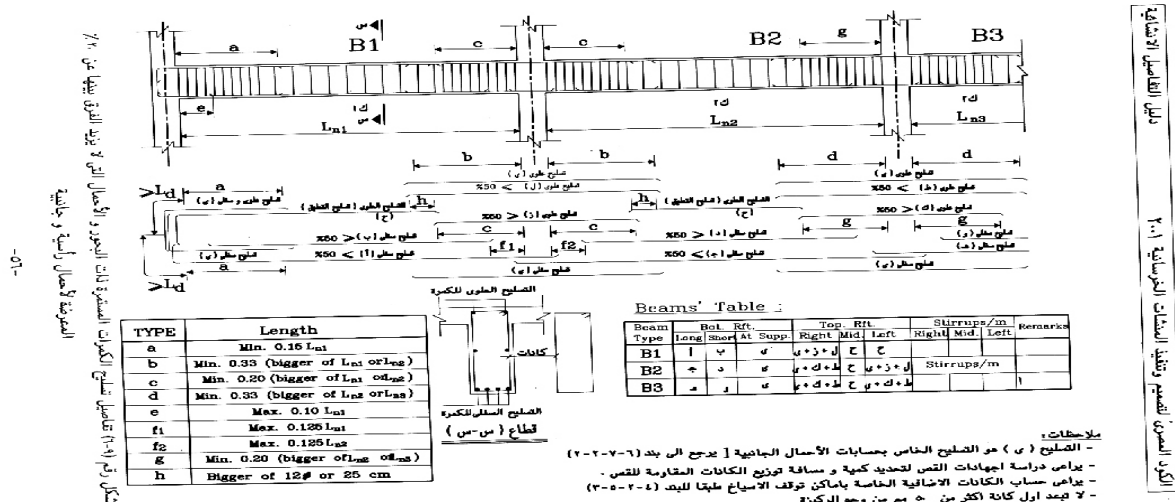
Wall – B02-W2



4 walls sections are consider unsafe.



C- Beams



- The additional seismic reinforcement is missing and shall be added.
- The beam straining actions resulting from the modifications increased, and the beams shall be redesigned accordingly.

D- Slabs

- The slabs straining actions resulting from the modifications increased , and the slabs shall be redesigned accordingly.



10. Conclusion

From above NECB recommend the following points are already done:

- 1- Check and modify model geometry
 - Verify storey heights, coordinates, offsets, slab/raft extents, stair flights and openings.
 - Correct any duplicated/wrongly-located nodes, duplicate elements, or misplaced supports.
- 2- Check for and add missing loads.
- 3- Increase the mentioned loads.
- 4- Modify seismic parameters
- 5- Run analysis and check vertical elements after modifications
 - **Based on the above (Point No. 09: Structural Models Output Review), NECB concludes that the superstructure is considered unsafe and requires redesign. After final confirmation, the design procedure will be as follows::**
 - 1- Create a 3-D foundation model (with $R = 3$)
 - 2- Update raft model with new loads
 - Check bearing pressures, Punching/shear and reinforcement .

11. Attachment

- 1- Coordination drawings.

